



**MASTERMIND
INDIA**

Spirit of India Special

RUNNING ON EMPTY

When Puli Ravi Kumar, 42, assistant professor at the National Institute of Technology (NIT), Warangal, discussed his dream project of making a car that would run on air with the director of the institute, he was greeted with a mixture of bafflement and boredom. But support came from an unexpected corner. Six final year B.Tech students of the Christu Jyoti Institute of Technology & Sciences (CJITS) in Warangal—N. Joy Kumar, 26, 23-year-olds M. Koushik, Y. Vasanth, P. Deepak Kumar, and G. Rajesh and 22-year-old R. Purnachandar Reddy—wanted to do their mandatory course project on his proposed air car. The professor was more than happy to oblige. Nine months later, Kumar and his team came up with an innovation where they showed that it was possible to run a car with just air without using petrol or diesel.

Their effort was aimed at bringing air pressure technology into commercial use in order to have a cleaner environment. "Our invention focuses on conversion of air pressure energy into mechanical force to run the car," says team leader Joy Kumar. The project was funded by Father K. Vijay Kumar Reddy, director of CJITS who gave the students Rs 50,000 for modifications in the cars besides arranging an old 118 Fiat diesel car and a petrol-fuelled Maruti 800.



(FROM LEFT) REDDY, RAVI KUMAR AND JOY KUMAR WITH THEIR AIR CARS

Puli RAVI KUMAR, 42

INNOVATION

Converting a fuel-driven car into one that runs on air

COST OF PRODUCTION

Rs 50,000

WHAT IT DOES

The kit allows an existing fuel-driven car to run only on air

WHILE OTHER AIR CAR MODELS RUN WITH AN AIR ENGINE, THIS ONE IS A MODIFICATION OF EXISTING ENGINES.

The air car is based on the principle of sending pressurised air into a cylinder which gives the piston movement which in turn runs the motor. The vehicle does not have a radiator as it does not involve any combustion. The best part is that it is totally pollution-free. They applied for a patent in Chennai in May 2009.

Though some French engineers had earlier made a prototype of the air car, Ravi Kumar's version is special. While other prototypes run with an air engine, this model involves the modification of existing engines. This is why it has tremendous commercial potential. However, due to some limitations on speed and availability of high pressure air, the commercial production of the air kit has not been possible yet. Initially, they introduced the air from the inlet valve. Now, they are making new air injectors, air injection pumps and throttle valves for supply of air according to the acceleration.

While five of the original team have gone abroad for further studies, Joy Kumar is working with Ravi Kumar to develop a full-fledged commercial model at NIT. They hope to offer a free ride soon.

by Amarnath K. Menon